PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	A Systematic Review Protocol to Assess Artificial Intelligence Diagnostic Accuracy Performance in Detecting Acute Ischemic Stroke and Large Vessel Occlusions on CT and MR Medical Imaging
AUTHORS	Kundeti, Srinivasa Rao; Vaidyanathan, Manikanda Krishnan; Shivashankar, Bharath; Gorthi, Sankar Prasad

VERSION 1 – REVIEW

REVIEWER	Gilbert Lim National University of Singapore, Singapore The reviewer is co-founder of a firm involved in the detection of diseases from retinal images.
REVIEW RETURNED	24-Aug-2020

GENERAL COMMENTS	The authors have proposed a systematic review protocol for assessing artificial intelligence (AI) diagnostic accuracy, for detecting ischemic stroke (IS) & large vessel occlusions (LVO) through medical imaging. The protocol appears broadly sound, and the motivation for a systematic review on medical imaging in stroke detection is well-founded and timely. However, there may remain some points for consideration: 1. While the exact combination of modality (imaging), method (AI) and diseases (IS/LVO) appears unique, there have been a number of recent studies that may be considered to overlap significantly with this scope. "Artificial intelligence to diagnose ischemic stroke and identify large vessel occlusions: a systematic review", Murray et al. (2020) is cited as [12] but only very briefly discussed, and a brief search produces more papers such as "Perfusion CT for prediction of hemorrhagic transformation in acute ischemic stroke: a systematic review and meta-analysis", Suh et al. (2019), "Using deep learning techniques in medical imaging: a systematic review of applications on CT and PET", Domingues et al. (2020) and "Artificial intelligence to diagnose ischemic stroke and identify large vessel occlusions: a systematic review", Murray et al. (2020). The authors might thus consider discussing such prior work in greater detail, and emphasize how the proposed protocol differs from them. 2. Further on the protocol, although the title purports to assess studies using medical imaging, the modalities involved appear to be limited to MR and CT images (Page 2, Line 55; Table 1; Candidate predictors in Table 3). The authors might consider defining the scope more precisely in the title, or expanding the scope to include

	other reasonable imaging modalities. 3. Certain abbreviations (e.g. MRI) might be defined before first use. 4. The relationship between ischemic stroke and LVO might be discussed in greater detail. In particular, are they to be considered equivalent outcomes, or will they be analyzed separately? 5. The search strategy methodology as presented in Table 2 might be specified in greater detail, e.g. are the search terms for the article title only, or article and text (this might be standard for the given online bibliographic databases, but it would be nice to have it explicitly stated)? 6. For Table 2, does the asterix before the "patient", "subject" and "scan" terms for search term group #4 have any special meaning? 7. On the diagnosis of ischemic stroke, it is both stated that "healthcare professional consensus or the standard-of-care diagnosis", and the "Alberta stroke program early CT score (ASPECTS) or Infarct volume on CT and MR scans" will be used (Page 4, Lines 44-47). The authors might clarify as to the procedure, were a paper to report performance based on some standard that is not ASPECTS/Infarct volume. Would the results be converted to some fixed standard if possible, or the paper excluded from the review? If no fixed standard is established, how would the possible meta-analysis be conducted? 8. There remain a number of spelling issues (e.g. "strokes" instead of "stroke", "backbox" instead of "blackbox", "...review is a summary..." instead of "...review in a summary..." on Page 1), which might be rectified. The authors might also consider having the manuscript further proofread for style (e.g. on use of capitalization).

REVIEWER	Philip Hoelter, MD Neuroradiology Department, University Erlangen-Nürnberg
REVIEW RETURNED	03-Sep-2020

GENERAL COMMENTS	The authors want to review current literature on AI based stroke diagnostic. First, I think this is a topic of great interest, as AI based image analysis is continuously increasing and more and more complements radiological image assessment. Stroke diagnosis might be challenging especially for non-trained professionals, as changes of the parenchyma and distal vasculature might be subtle. Anyway, I have to address some issues to the planned review: The authors state, that their objective is to evaluate diagnostic performance of acute ischemic stroke (AIS) diagnosis and detection of large vessel occlusion. How exactly do the authors wan to define “diagnosis of stroke”? They state that they want to do that based on the ASPECTS. That is not correct! The ASPECTS is designed to estimate parenchymal damage. An ASPECTS of 10 does not mean that the patient has no stroke! It means that if the patient has a stroke, there is lots of parenchyma to save! To detect AIS, ASPECTS is not a correct choice. Moreover, the authors state that they use ASPECTS or infarct volume to detect AIS. With regard to infarct volume, the authors

	should provide a more detailed definition on how they define infarct volume. Do they use NECT, CTP, MRI? For every imaging modality a definition of the volume is needed, as the calculations based upon the modalities are not necessarily comparable. This leads to another issue: the source data. I have to stress out that the source data used in each study will have an impact on the results! The authors need to consider those when comparing different studies. For instance, when checking CTP data, the scanning protocol has an enormous effect on the calculated results. Same for the planned detection of collaterals. How do you define the collateral status? There are plenty of assessment tools for collateral imaging. Will you consider CTA, 4D-CTA or CTP derived collateral status? How do you want to compare those? With regard to LVO detection, you need to explain what kind of LVO detection you want to compare (based on CTA, 4D-CTA, FDCT, CTP, Tof-MRI, CE-MRI….). The authors must step up efforts to explain what they really want to compare. They need to give a more detailed report on the different imaging modalities which AI software is applied on. Moreover, the term “stroke diagnosis” based on ASPECTS is wrong. Please rephrase that. Acute ischemic stroke diagnosis is based on two to three columns: NECT, CTA and (if applicable) CTP. I think that time window is an important factor that was not considered at all. However, the importance of the time window is reflected by current guidelines of the AHA/ASA. Please state on that. Furthermore, you could extent the reference section by more current literature.

REVIEWER	Kiruba Nagaratnam Royal Berkshire NHS Foundation Trust, Reading, UK
REVIEW RETURNED	05-Sep-2020

GENERAL COMMENTS	It should be clearly defined that the AI in stroke is narrow artificial intelligence and is used as a clinical decision support tool (CDS) rather than a diagnostic tool. Limitations of this review should be included. Please consider the following 1. The review will not include the non-diagnostic benefits of AI in stroke pathways such as improving clinicians' confidence in diagnosing ischaemic stroke/LVO with the help of AI clinical decision support tool, increasing the number of patients undergoing reperfusion treatments (thrombolysis or mechanical thrombectomy), 2. AI algorithms are dynamic. So their performance likely to improve with 'learning' from expanding data with time. So the conclusions on the basis of past performance may not be conclusive 3. Lack of head to head trials comparing AI products or various algorithms/combinations 4. Lack of information on source data upon which the AI algorithms are derived from. This constitutes an inherent bias on the performance of AI products in clinical trials/real life 5. Penumbra imaging (CT/MR perfusion) is an important area in stroke care where AI has been increasingly deployed. This has not been included in the analysis Please consider the following in the outcome 1. Clearly define detecting presence or absence of ischaemic stroke and ASPECTS scoring as two separate outcomes. Although they are interdependent in clinical practice they require different levels of skills.

	Language and writing - important for the quality 1. Consider rephrasing the opening sentence of the abstract e.g. Timely detection and treatment decision are crucial to the management of acute ischaemic stroke. Any delay in diagnosis or treatment can result in poor outcome.... 2. Check the manuscript to remove capitalisation of words 'clinicians' (page 4, line 41), 'infarct' (page 4, line 55), 'quality' (page 5, line5), 'stroke' (page 5, line 21), 'eight' (page 5 line 45), 'diagnostic' (page 6, line36), sensitivity, specificity, confidence interval (page 8 lines 52&53) 2. Capitalise the words appropriately - RCT, ASPECTS in the relevant paragraphs 3. Spelling error in Table 1 - Outcome row - 'diagnose' 4. Check the manuscript thoroughly for similar errors 4. What do you mean by 'one significant constraint is that the detection of ischaemic stroke and assessing LVO doesn't report results using different methods and outcome measures to detect or quantify the outcomes, which may lead to more optimistic results with overfitting'?

VERSION 1 – AUTHOR RESPONSE

Reviewer	Sl.No	Reviewer Comments	Addressed Line Numbers	Addressed?	Authors Remarks
Reviewer 1 Reviewer Name: Gilbert Lim Institution and Country: National University of Singapore, Singapore	1	While the exact combination of modality (imaging), method (AI) and diseases (IS/LVO) appears unique, there have been a number of recent studies that may be considered to overlap significantly with this scope. "Artificial intelligence to diagnose ischemic stroke and identify large vessel occlusions: a systematic review", Murray et al. (2020) is cited as [12] but only very briefly discussed, and a brief search produces more papers such as "Perfusion CT for prediction of hemorrhagic transformation in acute ischemic stroke: a systematic review and meta-analysis", Suh et al. (2019), "Using deep learning techniques in medical imaging: a systematic review of applications on CT and PET", Domingues et al. (2020) and "Artificial intelligence to diagnose	143-202	Addressed	Thank you for posing this valid question. We dicussed all the papers in depth, initially summarizing their methods and findinds. Atlast, we differentiated from Our protocol. Suh et al. (2019): addressed Hemmorahagic and CT perfusion Domingues et al. (2020): addresssed various DL models in CT and PET. Mostly narrative style Murray et al. (2020): is more closer to our work, as we mentioned in our paper. but lacks

	ischemic stroke and identify large vessel occlusions: a systematic review", Murray et al. (2020).			in use of systematiic review approach - like Risk of bias, strong inclusion/exclusoi n criteria - like internal and external validation
2	Further on the protocol, although the title purports to assess studies using medical imaging, the modalities involved appear to be limited to MR and CT images (Page 2, Line 55; Table 1; Candidate predictors in Table 3). The authors might consider defining the scope more precisely in the title, or expanding the scope to include other reasonable imaging modalities.	4-5; 242-245	Addressed	Modified the title and Incl/excl criteria and scope only to MR and CT medical imaging. Title & table 1 intervention row is changed
3	Certain abbreviations (e.g. MRI) might be defined before first use	Entire Document	Addressed	All abbrevations expanded on first use
4	The relationship between ischemic stroke and LVO might be discussed in greater detail. In particular, are they to be considered equivalent outcomes, or will they be analyzed separately?	255-295	Addressed	Yes more details provided. We added a new table - table 2 and explained in great detail on each of the outcome
5	The search strategy methodology as presented in Table 2 might be specified in greater detail, e.g. are the search terms for the article title only, or article and text (this might be standard for the given online bibliographic databases, but it would be nice to have it explicitly stated)?	330-332	Addressed	Table 2 has become table 3 now. In table3, we used title/abstract standard format
6	For Table 2, does the asterix before the "patient", "subject" and "scan" terms for search term group #4 have any special meaning?	324-327	Addressed	Explained the meaning of * in the paper. We used wild card to avoid missing plurals
7	On the diagnosis of ischemic stroke, it is both stated that "healthcare professional consensus or the standard-of-care diagnosis", and the "Alberta stroke program early CT score (ASPECTS) or Infarct volume on CT and	246-295	Addressed	Yes, the stand measures of out come will be compared between AI models and clinicians, where Some times

		MR scans" will be used (Page 4, Lines 44-47). The authors might clarify as to the procedure, were a paper to report performance based on some standard that is not ASPECTS/Infarct volume. Would the results be converted to some fixed standard if possible, or the paper excluded from the review? If no fixed standard is established, how would the possible meta-analysis be conducted?			clinicians will be the reference standard for AI model, in other cases, there will be a comparison given among clinicians which shall be reference standard for both cliicians and AI models. Sometimes, other models will be used as a refernce standard. so, based on our analysis , we will use these reference standards seperately to do our analysis. In case where AI models are just compared with other models, where there in no base human reference standard, we shall see if this model is an industry standard(e.g. like RAPID). If not, we will exlcude these from our Systematic review. briefly described this in the paper.
	8	There remain a number of spelling issues (e.g. "strokes" instead of "stroke", "backbox" instead of "blackbox", "...review is a summary..." instead of "...review in a summary..." on Page 1), which might be rectified. The authors might also consider having the manuscript further proofread for style (e.g. on use of capitalization).	Entire Document	Addressed	All Spelliing mistakes addressed and proof read
Reviewer 2					
Reviewer	1	The authors want to review current literature on AI based stroke diagnostic.	88-130	Addressed	Addded a few more literature aspects on

Name: Philip Hoelter, MD Institution and Country: Neuroradiology Department, University Erlangen-Nürnberg, Germany					diagnosis of ischemic stroke and importance of tissue window and brain imaging. Re-inforced with AHA/ASA standards
	2	The authors state, that their objective is to evaluate diagnostic performance of acute ischemic stroke (AIS) diagnosis and detection of large vessel occlusion. How exactly do the authors want to define “diagnosis of stroke”? They state that they want to do that based on the ASPECTS. That is not correct! The ASPECTS is designed to estimate parenchymal damage. An ASPECTS of 10 does not mean that the patient has no stroke! It means that if the patient has a stroke, there is lots of parenchyma to save! To detect AIS, ASPECTS is not a correct choice. Moreover, the authors state that they use ASPECTS or infarct volume to detect AIS. With regard to infarct volume, the authors should provide a more detailed definition on how they define infarct volume. Do they use NECT, CTP, MRI? For every imaging modality a definition of the volume is needed, as the calculations based upon the modalities are not necessarily comparable.	246-305	Addressed	Thank you for posing this valid question. Created a **new Table 2** and provided a clear mapping of Outcomes to Common brain Imaging used and how AI models work on sub-usecases in detection of AIS. Refer newly added table 2.
	3	This leads to another issue: the source data. I have to stress out that the source data used in each study will have an impact on the results! The authors need to consider those when comparing different studies. For instance, when checking CTP data, the scanning protocol has an enormous effect on the calculated results. Same for the planned detection of collaterals. How do you	246-305	Addressed	Yes. Different models use different Source data, approaches and validatioin methodologies which might leadd to information. We explained to tackle this by having expert reviews, looking at the sample sizes, georgraphies and

	define the collateral status? There are plenty of assessment tools for collateral imaging. Will you consider CTA, 4D-CTA or CTP derived collateral status? How do you want to compare those?			validation methods and subgrouping. Explained in detailed in the paper.
4	With regard to LVO detection, you need to explain what kind of LVO detection you want to compare (based on CTA, 4D-CTA, FDCT, CTP, Tof-MRI, CE-MRI….). The authors must step up efforts to explain what they really want to compare. They need to give a more detailed report on the different imaging modalities which AI software is applied on. Moreover, the term “stroke diagnosis” based on ASPECTS is wrong. Please rephrase that. Acute ischemic stroke diagnosis is based on two to three columns: NECT, CTA and (if applicable) CTP.	246-305	Addressed	Please find explanation in-detail the protocol and also in **Table 2.** Created a new Table 2 and provided a clear mapping Mainly our Protocol does not include Perfusion imaging (CTP and MRP) - as this is optional.
5	I think that time window is an important factor that was not considered at all. However, the importance of the time window is reflected by current guidelines of the AHA/ASA. Please state on that.	107-130	Addressed	Regarding the time window and AHA/ASA standard updates are made in the **Background section**. Also established that time window and tissue window are complementary and this protocol scope is to only focus on tissue window.
6	Furthermore, you could extent the reference section by more current literature.	99-149	Addressed	Please refer the updated Background section and updated references section. Given sufficient literature on Stroke practices, standards, AI imaging and others.

Reviewer3 Reviewer Name: Kiruba Nagaratnam Institution and Country: Royal Berkshire NHS Foundation Trust, Reading, UK	1	Limitations of this review should be included. Please consider the following 1. The review will not include the non-diagnostic benefits of AI in stroke pathways such as improving clinicians' confidence in diagnosing ischaemic stroke/LVO with the help of AI clinical decision support tool, increasing the number of patients undergoing reperfusion treatments (thrombolysis or mechanical thrombectomy), 2. AI algorithms are dynamic. So their performance likely to improve with 'learning' from expanding data with time. So the conclusions on the basis of past performance may not be conclusive 3. Lack of head to head trials comparing AI products or various algorithms/combinations 4. Lack of information on source data upon which the AI algorithms are derived from. This constitutes an inherent bias on the performance of AI products in clinical trials/real life 5. Penumbra imaging (CT/MR perfusion) is an important area in stroke care where AI has been increasingly deployed. This has not been included in the analysis	421-445; 60-73	Addressed	Thank you for providing the valuable input. We kept some of these points for our Systematic review paper to thoroughly check on these points - like how AI models are comparing and establishing the performance metrics with Clinicians, etc. After your suggestion, we felt that these limitations are important to mention upfront in protocol. so, Updated the Discussion and Conclusion section with these Limitatioons. Yes - Penumbra imaging is not part of the review , as its not class I level evidence. We already have CTA that capures the necessary hemodynamic information.
	2	Please consider the following in the outcome 1. Clearly define detecting presence or absence of ischaemic stroke and ASPECTS scoring as two separate outcomes. Although they are interdependent in clinical practice they require different levels of skills.	298-305	Addressed	Yes. Thank you we have considererd is - similar to reviewer 2 request. Updated the outcomes and relatively how they are implemented on AI in table 2 and discussed in detail

3	Language and writing - important for the quality 1. Consider rephrasing the opening sentence of the abstract e.g. Timely detection and treatment decision are crucial to the management of acute ischaemic stroke. Any delay in diagnosis or treatment can result in poor outcome.... 2. Check the manuscript to remove capitalisation of words 'clinicians' (page 4, line 41), 'infarct' (page 4, line 55), 'quality' (page 5, line5), 'stroke' (page 5, line 21), 'eight' (page 5 line 45), 'diagnostic' (page 6, line36), sensitivity, specificity, confidence interval (page 8 lines 52&53) 2. Capitalise the words appropriately - RCT, ASPECTS in the relevant paragraphs 3. Spelling error in Table 1 - Outcome row - 'diagnose' 4. Check the manuscript thoroughly for similar errors	Entire Document	Addressed	Yes. Thank you for pointing these spelling and punctuation isssues. We have rephreased opening sentence in abstract and also took care of all similar errors throught out the ddocument. We also have taken help of professional writing team.
4	What do you mean by 'one significant constraint is that the detection of ischaemic stroke and assessing LVO doesn't report results using different methods and outcome measures to detect or quantify the outcomes, which may lead to more optimistic results with overfitting'?	433-436	Addressed	Rephrased in the mentioned line numbers. Thank you for pointing this out

5	FORMATTING AMENDMENTS (if any) Required amendments will be listed here; please include these changes in your revised version: ● Patient and Public Involvement: Authors must include a statement in the methods section of the manuscript under the sub-heading 'Patient and Public Involvement'. This should provide a brief response to the following questions: How was the development of the research question and outcome measures informed by patients' priorities, experience, and preferences? How did you involve patients in the design of this study? Were patients involved in the recruitment to and conduct of the study? How will the results be disseminated to study participants? For randomised controlled trials, was the burden of the intervention assessed by patients themselves? Patient advisers should also be thanked in the contributorship statement/acknowledgements. If there is no patient involved in the study, please state "No patient involved" under the sub-heading 'Patient and public involvement'.	496	Addressed	Updated this in the protocol: Patient and public involvement: No patient involved

VERSION 2 – REVIEW

REVIEWER	Gilbert Lim Singapore Eye Research Institute Singapore
REVIEW RETURNED	22-Oct-2020

GENERAL COMMENTS	We thank the authors for largely addressing the major concerns

	raised from the last review round. Some minor spelling/grammatical issues remain (e.g. "noncontrast computed tomographyy", Line 120; "lack of [a] fixed reference standard", Line 250; "clinician's performance" -> "clinicians' performance", Line 252; "Perfusion studies [are] not in the scope...", Line 268; etc.)

REVIEWER	Philip Hoelter Neuroradiology University Hospital Erlangen
REVIEW RETURNED	09-Nov-2020

GENERAL COMMENTS	First, thank you for revising your manuscript. The study aims to assess the accuracy of AIS diagnosis when performed by AI frameworks. I do have some minor remarks: INRTRODUCTION Background Line 102 ff. “LVO variability…” The site of occlusion, LVO occurrence respectively, is not inevitable linked to the cause of stroke. For instance, a cardioembolic stroke may occur in the posterior circulation as well. Please correct. Line 104 ff. “patients with AIS…” Patients with LVO of the posterior circulation are qualified for EVt as well. Please correct. Line 121f. “Generally, …” The time window is more and more replaced by the tissue window. Please state on that. MATERLIAS AND METHODS Inclusion and exclusion criteria Line 350f. “Detecting LVO outcome..” Generally, the term “outcome” is used for the outcome of the patient, as measured by the mRS. I don’t know what you mean by LVO outcome and CBS. If you mean the clot extend, than please write that. Please state on that. With regards to mismatch score prediction, you might have to deal with different softwares and different core/penumbra prediction models. Maybe you should state on that within the text.

REVIEWER	Kiruba Nagaratnam Royal Berkshire NHS Foundation Trust, Reading. UK
REVIEW RETURNED	10-Nov-2020

GENERAL COMMENTS	Thank you for considering the comments and for taking them into account on your revised manuscript. I have read the revised manuscript thoroughly. I am afraid to say the revised manuscript appears less focused than the first one. It is good to see you have included the papers the reviewers have suggested. But it appears you have failed to use them in a manner which strengthens the case for your study. This leaves the reader unclear about the objectives and how you are planning to measure them. You have included my suggestions about the potential limitation. But it appears the comments have been copied and pasted. You have mentioned in your response that professional writers have reviewed the manuscript. But I am afraid, in my view the writing does not appear to be of sufficient standard expected for a scientific communication.

VERSION 2 – AUTHOR RESPONSE

Reviewer	Sl.No	Reviewer Comments	Addressed Line Numbers	Addressed?	Authors Remarks
Reviewer: 1 Reviewer Name: Gilbert Lim	1	We thank the authors for largely addressing the major concerns raised from the last review round. Some minor spelling/grammatical issues remain (e.g. "noncontrast computed tomographyy", Line 120; "lack of [a] fixed reference standard", Line 250; "clinician's performance" -> "clinicians' performance", Line 252; "Perfusion studies [are] not in the scope...", Line 268; etc.)	134; 295-296; 297; 316	yes	Thank you for showing a few spelling/grammatical issues. We thoroughly reviewed to correct all spelling mistakes.

Reviewer: 2 Reviewer Name: Philip Hoelter	1	First, thank you for revising your manuscript. The study aims to assess the accuracy of AIS diagnosis when performed by AI frameworks. I do have some minor remarks: INRTRODUCTION Background Line 102 ff. "LVO variability…" The site of occlusion, LVO occurrence respectively, is not inevitable linked to the cause of stroke. For instance, a cardioembolic stroke may occur in the posterior circulation as well. Please correct. Line 104 ff. "patients with AIS…" Patients with LVO of the posterior circulation are qualified for EVt as well. Please correct. Line 121f. "Generally, …" The time window is more and more replaced by the tissue window. Please state on that.	90-91; 93-95;112-115	yes	Thank you for the minor remarks. Thank you for your suggestion, we modified the same

	2	MATERLIAS AND METHODS Inclusion and exclusion criteria Line 350f. “Detecting LVO outcome..” Generally, the term “outcome” is used for the outcome of the patient, as measured by the mRS. I don’t know what you mean by LVO outcome and CBS. If you mean the clot extend, than please write that. Please state on that. With regards to mismatch score prediction, you might have to deal with different softwares and different core/penumbra prediction models. Maybe you should state on that within the text.	300-308, 313-323, 336-338 ;336-337; 73-74, 504-506	yes	Yes. We removed outcome from PICO outcomes as suggested by you. (300-308, 313-323, 336-338) We mean clot extend, as you pointed and explained the same (336-337) As perfusion is optional and our limited scope , mismatch score prediction is out-of-scope in this review(73-74, 504-506)
Reviewer: 3 Reviewer Name: Kiruba Nagaratnam	1	Thank you for considering the comments and for taking them into account on your revised manuscript. I have read the revised manuscript thoroughly. I am afraid to say the revised manuscript appears less focused than the first one. It is good to see you have included the	174-236	yes	Thank you for your critical and kind review. In the revised manuscript, we also tried to shortened and concise points are made in Background section to motivate the reader towards the objectives targeted by the paper.

		papers the reviewers have suggested. But it appears you have failed to use them in a manner which strengthens the case for your study. This leaves the reader unclear about the objectives and how you are planning to measure them.			
	2	You have included my suggestions about the potential limitation. But it appears the comments have been copied and pasted. You have mentioned in your response that professional writers have reviewed the manuscript. But I am afraid, in my view the writing does not appear to be of sufficient standard expected for a scientific communication.	500-522	yes	Thank you for your critical and kind review pointing out on AI model limitations. We too felt that these limitations are existing and would be nice to quote. Addition ally, now in this revision we had provided rationale in some places and explained in other places to make the reader aware of the challenges faced in this study context. Now we have strengthened these with reasons in "DISCUSSION line numbers 500 to 522". We have given this paper to our University scientific press, that has professionally reviewed, corre cted this paper. We learnt a lot in this process

					and thank you for challenging us.